

ECO-BIN

Project report submitted in partial fulfilment of the requirements for the award of the degree of

Bachelor of Science in Computer Science

Submitted by

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CERTIFICATE

This is to certify that report entitled “**Stay Easy System**” is a bonafide report of the project (CS1644MajorProject) presented during the Sixth Semester by **Akash G Anil (Reg.No:32021135007)**, **Allen Thomas (Reg.No: 32021135009)**, **Santhina S Nair (Reg.No:32021135023)**, **Sudhi Babu (Reg.No: 32021135025)** in partial fulfilment of the requirements for the award of the degree of **B.Sc. in Computer Science**.

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ABSTRACT

EcoBin is a Plastic Waste Collection Website that brings together the realms of waste management and digitalized online platform. This abstract introduces the key features and benefits of EcoBin, a platform designed to foster eco-consciousness and convenience for users.

The waste collection aspect of EcoBin aims to simplify and optimize waste disposal processes. Users can effortlessly schedule waste pickups, and receive timely notifications for collection timings. By offering a user-friendly interface and efficient waste management solutions, EcoBin seeks to encourage responsible waste handling and contribute to a cleaner environment.

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CHAPTER 1

INTRODUCTION

1.1 Background of the work

Plastic waste collection refers to the process of collecting, transporting, and disposing of plastic waste materials generated by households, businesses, or industrial activities. The primary objective of plastic waste collection is to ensure proper management and disposal of plastic to protect the environment and public health. However, the most challenging part of this waste collection is lack of communication between the customer and the worker. These causes inflexibility, limited waste segregation, manual labour intensity, low public awareness, lack of monitoring, and environmental/health risks. Through this project we are trying to digitalize current plastic waste collection method and provide a new platform for them to work with. This web application platforms offer the potential for more efficient scheduling, increased convenience, and improved communication between waste generators and service providers.

1.2 Definition of the Problem

Although there is government plan such as suchitwa mission. mission which is responsible for the collection of plastic waste from household but this traditional system has problems in them. Some problems faced by traditional methods of plastic waste collection includes:

- **Lack of Flexibility:** Traditional methods have set collection schedules that may not cater to individual needs or changes in waste generation.
- **Limited Communication and Tracking:** Traditional methods lack effective communication channels between waste collectors and users, making it difficult to track service requests, report issues, or receive updates
- **Limited Waste Segregation:** Traditional methods often rely on waste generators to segregate waste at the source, which may not always be followed diligently.
- **Low Public Awareness and Participation:** Traditional methods often struggle to engage the public in waste management practices.

1.3 Scope and Objective of the Work

The scope of online plastic waste collection in the waste management sector is significant and diverse. Online plastic waste collection platforms offer convenience, efficiency, and improved communication between waste generators and service providers. They promote waste segregation, and extend services to underserved areas. Together, ECOBIN contribute to enhancing waste management practices, promoting sustainability, and elevating the overall customer experience in the sector.

1.4 Existing System and Drawbacks

1.4.1 AAKRI

DESCRIPTION

The system offers residential and commercial scrap removal services, ensuring efficient and convenient disposal of unwanted materials. Additionally, organizes environmental projects such as beach cleanups,

planting initiatives, and recycle drives, promoting sustainability and preserving the natural beauty of our surroundings.

LIMITATION

When it comes to system's user experience, one potential disadvantage is that their websites may lack credibility or trustworthiness, which can affect users' confidence in their services. Another drawback is that this organization is privately owned, limiting their reach and resources compared to public entities. Lastly, their services may be exclusive to the city of Kochi, limiting accessibility for individuals in other locations.

1.4.2 Waste Management

DESCRIPTION

Our system offers convenient residential waste pickup services, ensuring the hassle-free removal of waste from households. They also provide business waste pickup solutions to cater to the needs of commercial establishments. Additionally, they offer dumpster rental services for larger-scale waste disposal projects.

LIMITATION

One major disadvantage of their system is that their services are limited to a specific area, exclusively serving North America. Another drawback is scheduling issues, where customers may face difficulties in securing convenient pickup times. Being a private organization, their resources and reach may be limited compared to public entities. Additionally, their services may be perceived as expensive, and the lack of a feedback mechanism hinders customer communication and improvement opportunities.

1.4.3 DJUNK

DESCRIPTION

Their system is focused on dynamic recycling and promoting green living. They prioritize solid waste management, ensuring proper handling and disposal of waste materials. They also emphasize recycling, encouraging individuals and businesses to participate in sustainable practices. By prioritizing these initiatives, their system aims to contribute to a cleaner and more environmentally-friendly future.

LIMITATION

One significant disadvantage of their system is an unresponsive website, which can lead to frustration and inconvenience for users. Another drawback is that their services are limited only to the city of Kochi, limiting accessibility for individuals outside the area. Additionally, the system is not user-friendly, making it difficult for users to navigate and utilize its features. Moreover, there are ongoing issues with user registration and login processes, further hindering the overall user experience.

1.4.4 Waste Connection

DESCRIPTION

Their system offers a range of services to address waste management needs. They provide convenient home trash pickup services, ensuring the efficient removal of household waste. Additionally, they offer business waste collection solutions to cater to the needs of commercial establishments. Their system also provides dumpster rental services for larger-scale waste disposal projects. Furthermore, they offer valuable recycling information to educate and encourage individuals and businesses to participate in sustainable recycling practices.

LIMITATION

One disadvantage of their system is that their services are limited to the United States and Canada, restricting access for potential customers outside these regions. Another drawback is that waste collection services can be expensive, potentially impacting affordability for some users. Being a private organization, their reach and resources may be limited compared to public entities. Users may also experience slow loading times or encounter broken links on their website, affecting the overall user experience. Additionally, waste collection may only be available during limited hours, causing inconvenience for customers.

1.5 Proposed System and Advantages

ECOBIN plays a crucial role in improving the current waste management system, aligning with initiatives like Suchitwa Mission. It offers customers the flexibility to choose the date and time for waste collection, ensuring convenience and efficiency. The system provides detailed guidance on recycling various items and offers instructions on proper waste sorting techniques. Additionally, it offers valuable tips on waste reduction, including suggestions for minimizing single-use plastic, composting, and upcycling. **ECOBIN** incorporates a feedback mechanism, allowing customers to provide input and share their experiences.

1.5.1 Advantages

- **ECOBIN** offers several advantages that contribute to improving the current waste management system, aligning with initiatives like Suchitwa Mission. One of its key benefits is the convenience it provides to customers. Users can choose the date and time for waste collection according to their preferences, making the process more flexible and efficient.
- The system goes beyond just waste collection by providing valuable information on recycling. It guides users on which items are recyclable and how to properly recycle them, promoting responsible waste disposal practices. Moreover, the system offers guidance on waste sorting techniques, ensuring that waste is separated correctly to maximize recycling efforts and minimize environmental impact.
- A feedback mechanism is implemented within the system, allowing customers to provide input and suggestions, fostering continuous improvement and enhancing user engagement. Additionally, the system rewards customers with points and badges, creating a gamified experience that encourages active participation in waste management activities.

CHAPTER 2

FEASIBILITY STUDY

2.1 Introduction

The feasibility study concerns with the considerations made to verify whether the proposed system is feasible in all terms. Once an idea to develop software is put forward, the question that arises first will pertain to the feasibility study.

The following feasibility studies were conducted based on STAY EASY SYSTEM are:

1. Technical Feasibility
2. Economic Feasibility
3. Operational Feasibility

2.2 Technical Feasibility

Technical feasibility report assesses the viability and suitability of implementing a waste collection and e-commerce website. The purpose of this report is to evaluate the technical aspects of the proposed system and determine its feasibility in meeting the requirements of waste management and e-commerce operations.

1. Compatibility:

The waste collection and e-commerce website will be designed to be compatible with a wide range of operating systems, devices, and web browsers. By employing responsive design techniques, the website will ensure seamless accessibility and optimal user experience across different platforms.

2. Scalability:

The proposed website will be built with scalability in mind, accommodating the potential growth of the user base and increasing volumes of waste collection data. The underlying architecture will be designed to handle high traffic and enable future expansion through the addition of new features and integration with other systems.

3. Security:

Security will be a paramount consideration for the waste collection and e-commerce website. It will employ robust encryption protocols to safeguard sensitive user information, financial transactions, and confidential waste management data. Regular security audits and vulnerability assessments will be conducted to identify and address potential vulnerabilities.

4. Functionality:

The website will offer comprehensive functionality to manage waste collection schedules, customer preferences, and route optimization. Features like notifications, alerts, and reminders will be implemented to ensure efficient waste collection appointments. Real-time tracking and monitoring capabilities will enable effective waste management operations.

5. Integration:

Seamless integration with external systems will be a key feature of the waste collection and e-commerce website. This will include integration with payment gateways to facilitate online transactions, as well as integration with inventory management systems and CRM platforms to streamline operations and enhance efficiency.

6. User Interface (UI) and User Experience (UX):

The website will prioritize a user-friendly interface to enhance usability for waste management personnel and customers. Intuitive navigation, clear design elements, and responsive layouts will contribute to a positive user experience. A visually appealing and intuitive UI will ensure ease of use and customer satisfaction.

7. Reliability and Support:

The website will be built with reliability in mind, minimizing downtime and system failures. Regular maintenance, backups, and disaster recovery mechanisms will be implemented to ensure uninterrupted service. Prompt customer support and technical assistance will be provided to address any issues and maintain a smooth operation.

8. Data Management:

Efficient data management will be a core component of the website. It will securely store and manage waste collection data, customer information, and transaction records. Advanced data analytics capabilities will enable insights for optimizing waste management processes and making data-driven decisions.

Conclusion:

Based on the technical assessment conducted, the waste collection and e-commerce website demonstrate strong technical feasibility. The proposed system is compatible, scalable, secure, functional, and integrable. It prioritizes user experience, reliability, and efficient data management. These technical considerations ensure the website's suitability in meeting the requirements of waste collection and e-commerce operations.

2.3 Economic Feasibility

Economic feasibility report evaluates the financial viability of implementing a waste collection and e-commerce website. The objective is to determine whether the proposed system is economically feasible and can generate sufficient benefits to justify the associated costs.

Cost Analysis:

The cost analysis involves assessing the expenses associated with developing, implementing, and maintaining the waste collection and e-commerce website. This includes expenses related to website development, hosting, software licenses, infrastructure, security measures, and ongoing maintenance and support.

Benefits Analysis:

The benefits analysis focuses on identifying the potential economic advantages and returns resulting from the implementation of the website. These benefits may include increased operational efficiency, cost savings through optimized waste collection routes, reduced administrative overhead, improved customer satisfaction, and increased revenue from e-commerce transactions.

Return on Investment (ROI):

Calculating the ROI helps determine the financial viability of the project. It involves comparing the expected benefits and cost savings over a specific period against the initial and ongoing costs of implementing and maintaining the website. A positive ROI indicates the economic feasibility of the project, while a negative ROI may require reassessment or adjustment of the project scope.

Revenue Generation:

The e-commerce component of the website offers potential revenue streams through online transactions. Revenue can be generated from the sale of waste management products, additional services, and promotional offers. Forecasting potential revenue and estimating market demand will provide insights into the website's revenue-generating capabilities.

Cost Reduction:

The waste collection aspect of the website can contribute to cost reduction by optimizing collection routes, minimizing fuel consumption, and reducing labour requirements. Efficient scheduling and route planning can lead to cost savings through improved resource allocation and reduced operational expenses.

Cost-Benefit Analysis:

Performing a cost-benefit analysis involves comparing the costs associated with implementing and maintaining the website against the quantifiable benefits gained. This analysis provides a comprehensive view of the economic feasibility by considering both the costs and the benefits in monetary terms.

Conclusion:

Based on the economic feasibility analysis, the waste collection and e-commerce website demonstrate positive potential for generating economic benefits. The cost analysis, ROI calculation, revenue generation potential, and cost reduction aspects indicate that the implementation of the website is economically feasible. However, it is crucial to regularly monitor and evaluate the financial performance of the website to ensure ongoing sustainability and profitability.

2.4 Operational Feasibility

Operational feasibility report assesses the practicality and suitability of implementing a waste collection and e-commerce website. The objective is to determine whether the proposed system can be effectively integrated into existing operations and processes, and whether it can be successfully adopted by waste management personnel and customers.

1. Stakeholder Analysis:

Analysing the stakeholders involved in waste collection and e-commerce operations is crucial for understanding their needs, requirements, and potential challenges. This analysis helps identify any potential conflicts or issues that may arise during system implementation and operation.

2. Process Integration:

The waste collection and e-commerce website should seamlessly integrate with existing waste management processes, such as collection scheduling, route planning, and customer communication. The system should enhance operational efficiency, streamline workflows, and improve overall waste management processes.

3. Training and Adoption:

Ensuring that waste management personnel and customers can easily adapt to the new system is essential for operational feasibility. Adequate training and support should be provided to enable personnel to effectively use the system for waste collection and management activities. Additionally, the website should have a user-friendly interface to facilitate customer adoption and engagement.

4. Infrastructure and Resources:

Assessing the availability of necessary infrastructure and resources is crucial for operational feasibility. This includes evaluating the technological capabilities, such as internet connectivity and hardware requirements, to support the website's operation. Sufficient resources, including personnel, equipment, and inventory, should be allocated to handle waste collection and e-commerce activities effectively.

5. Scalability and Flexibility:

The operational feasibility of the website relies on its scalability and flexibility to adapt to changing needs and future growth. The system should accommodate an increasing volume of waste collection requests, handle growing user traffic, and support potential expansion to other locations or services.

6. Customer Experience:

Providing a positive customer experience is vital for operational feasibility. The website should be user-friendly, offering intuitive navigation, clear instructions, and responsive design. It should enable customers to easily schedule waste collection appointments, track progress, and engage with the system's features, fostering customer satisfaction and loyalty.

7. Regulatory Compliance:

Ensuring compliance with relevant waste management regulations and guidelines is essential for operational feasibility. The website should adhere to legal requirements, data protection regulations, and industry standards to maintain operational integrity and gain stakeholder trust.

Conclusion:

Based on the operational feasibility analysis, the waste collection and e-commerce website demonstrate practicality and suitability for implementation. The system's integration with existing processes, training and adoption strategies, available infrastructure and resources, scalability, customer experience considerations, and regulatory compliance contribute to its operational feasibility. However, ongoing monitoring, evaluation, and continuous improvement efforts are necessary to optimize operational efficiency and maintain alignment with evolving needs and regulations.

CHAPTER 3 REQUIREMENT ANALYSIS

3.1 Hardware and Software Requirement Specification

Hardware requirements

In hardware requirements, we require all those components which will provide us with the platform required for the project. The minimum hardware required for the project is as follows. RAM minimum of 1GB or above, Hard disk minimum of 10 GB, Processor of Intel dual Core i3. These all are the minimum requirement required for Our project. We want to make our computer to be to be used in any. Therefore, we have taken minimum configuration to large extent. 1 GB RAM is used so that we can execute our project at least possible RAM. 20 GB RAM is used because project takes Less space to be executed or stored. Therefore, minimum hard disk is used. Other enhancements are required to needs.

Software requirements

The software needed for the project are windows 8 or above, Visual Studio code, MySQL, HTML, CSS, JavaScript, SQL DATABASE, Xampp, PHP.

CHAPTER 4

DESIGN AND IMPLEMENTATION

4.1 DFD

A data-flow diagram is a way of representing a flow of data through a process or a system. The DFD also provides information about the outputs and inputs of each entity and the process itself. We have shown the dataflow of the system through the level 0, level 1, level 2 and level 3 DFD. Four simple notations are used to complete a DFD. The notations are given below: -

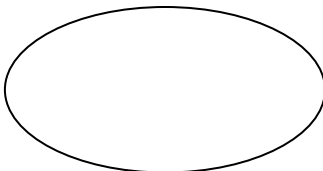
Data flow

The data flow is to describe the movement of information from one part of the system to another part. Flows represent data in motion. It's a pipe-line through which information flows. Data flow is represented by an arrow.



Process

A circle or bubble represents a process that transforms incoming data to outgoing data. Process shows a part of the system that transforms inputs to outputs.



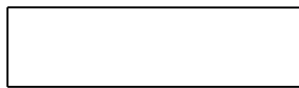
External Entity

A square defines a source or destination of a system data. External entities represent any entity that supplies or receive information from the system but is not part of the system.



Data Store

The data store represents a logical file. A logical file can represent either a data store symbol which can represent either a data structure or a physical file on disk. The data store is used to collect data at rest or temporary repository of data. It is represented by open rectangle



4.1.1 LEVEL -0 DFD (CONTEXT DIAGRAM)

It is known as fundamental system model, or context diagram represents the entire software requirement as a single bubble with input and output data denoted by incoming and outgoing arrows.

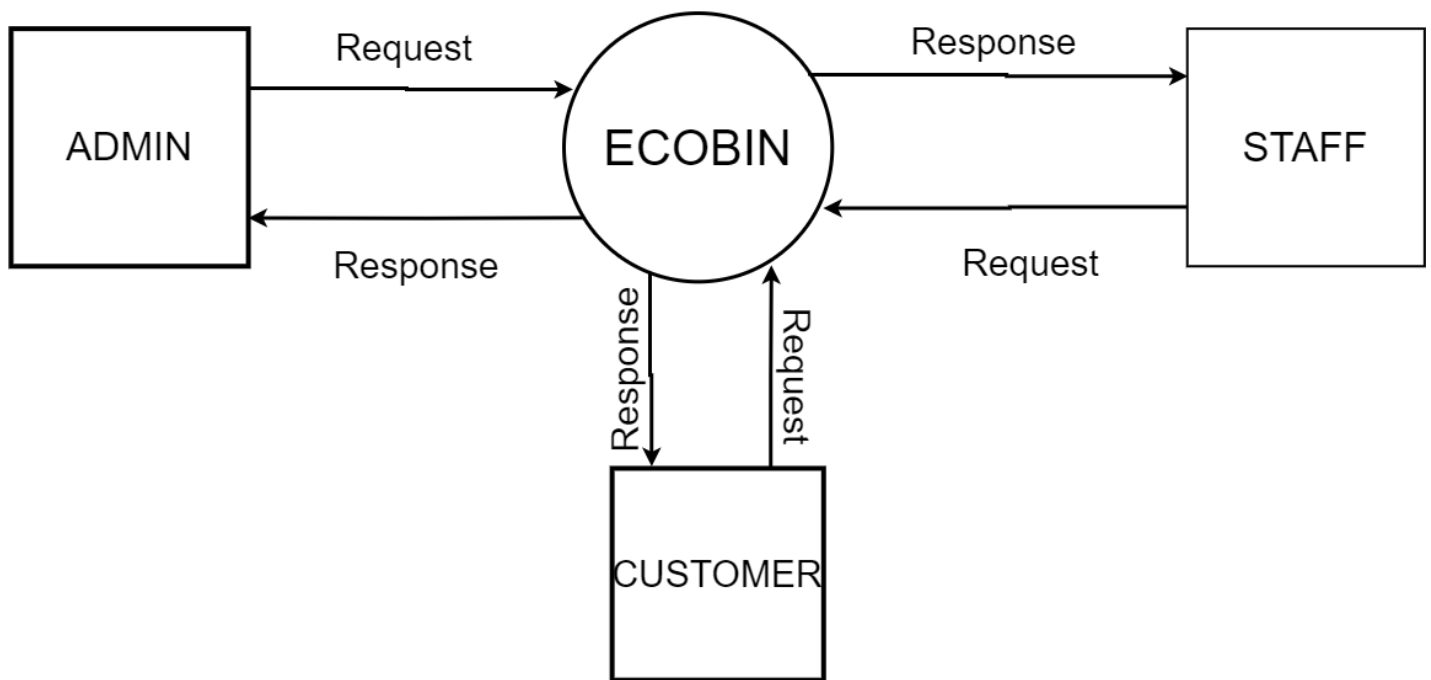


Image 4.1.1 Level 0 DFD

The ECOBIN is associated with admin, customer, staff, and delivery boy.

At the top level, The Admin user has full control over the system. They can manage settings, user permissions, and access reports and analytics.

The Staff, who performs operational tasks, interacts with the admin and carries out responsibilities such as waste collection, updating routes, and handling customer request.

The Customer user engages with the system to access waste collection services and make e-commerce transactions. They can register, log in, and set preferences for waste collection schedules and services. Customers can browse products, place orders, make online payments, and track their order status.

Finally, the Delivery Boy user receives assigned waste collection tasks and delivery orders from the system. They deliver products. The Delivery Boy updates the system to confirm successful order delivery.

4.1.2 LEVEL-1 DFD

In level-1 DFD, a context diagram is decomposed into multiple bubbles/processes. In this level, we highlight the main objectives of the system and breakdown the high-level process of level-0 DFD into sub processes.

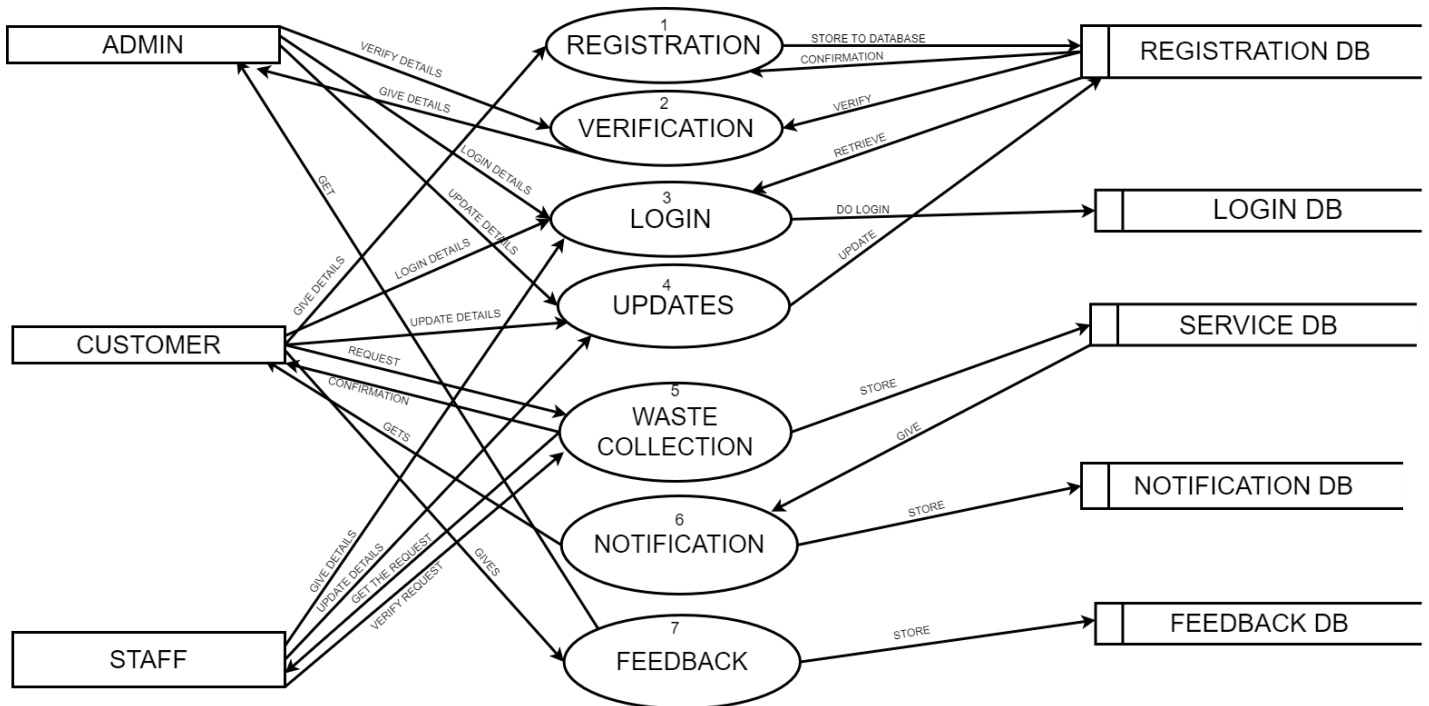


Image 4.1.2 Level 1 DFD

4.1.3 LEVEL-2 DFD

Level-2 DFD goes one process deeper into parts of level-1 DFD. It can be used to project the specific or necessary details about the system's functioning.

4.1.3.1 Waste Collection (5)

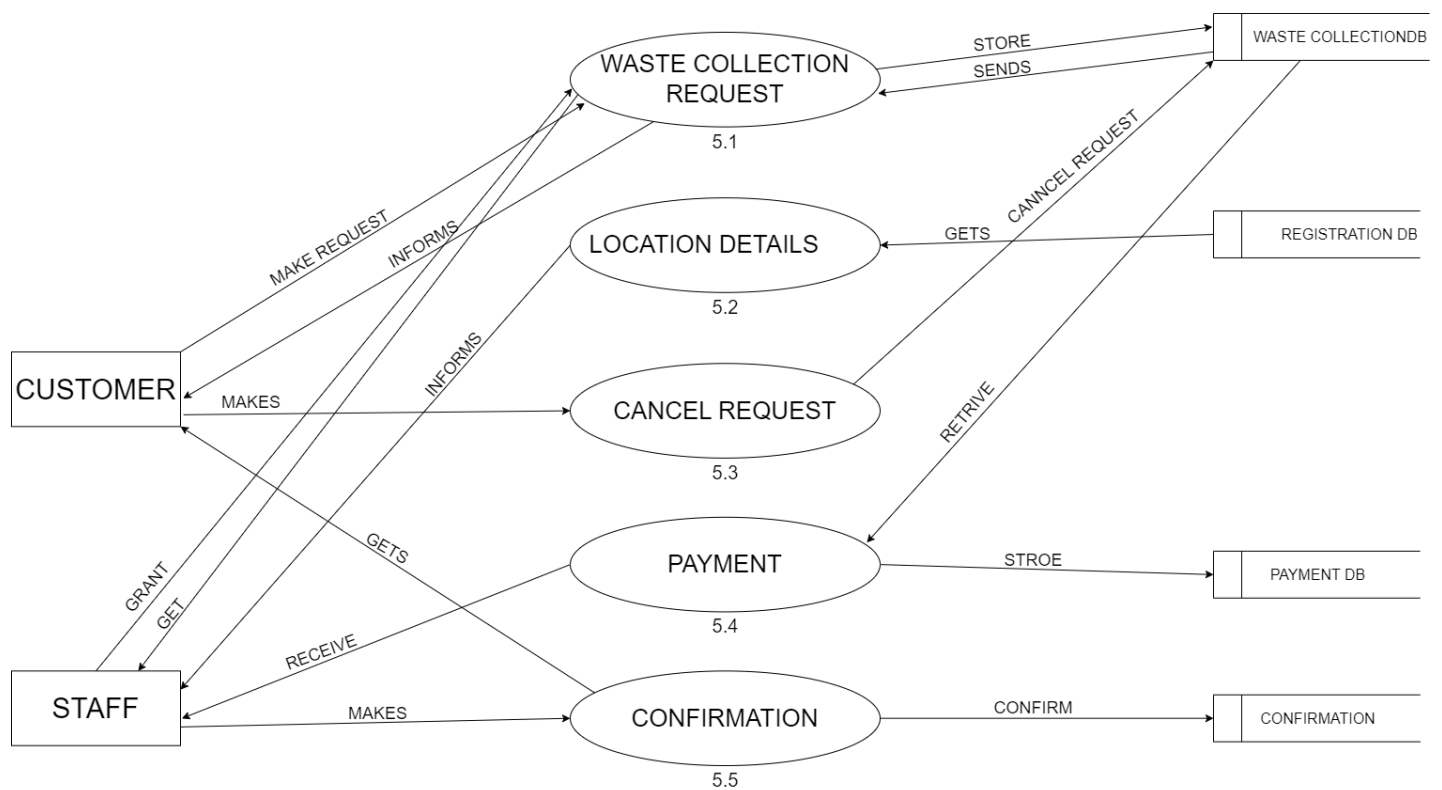
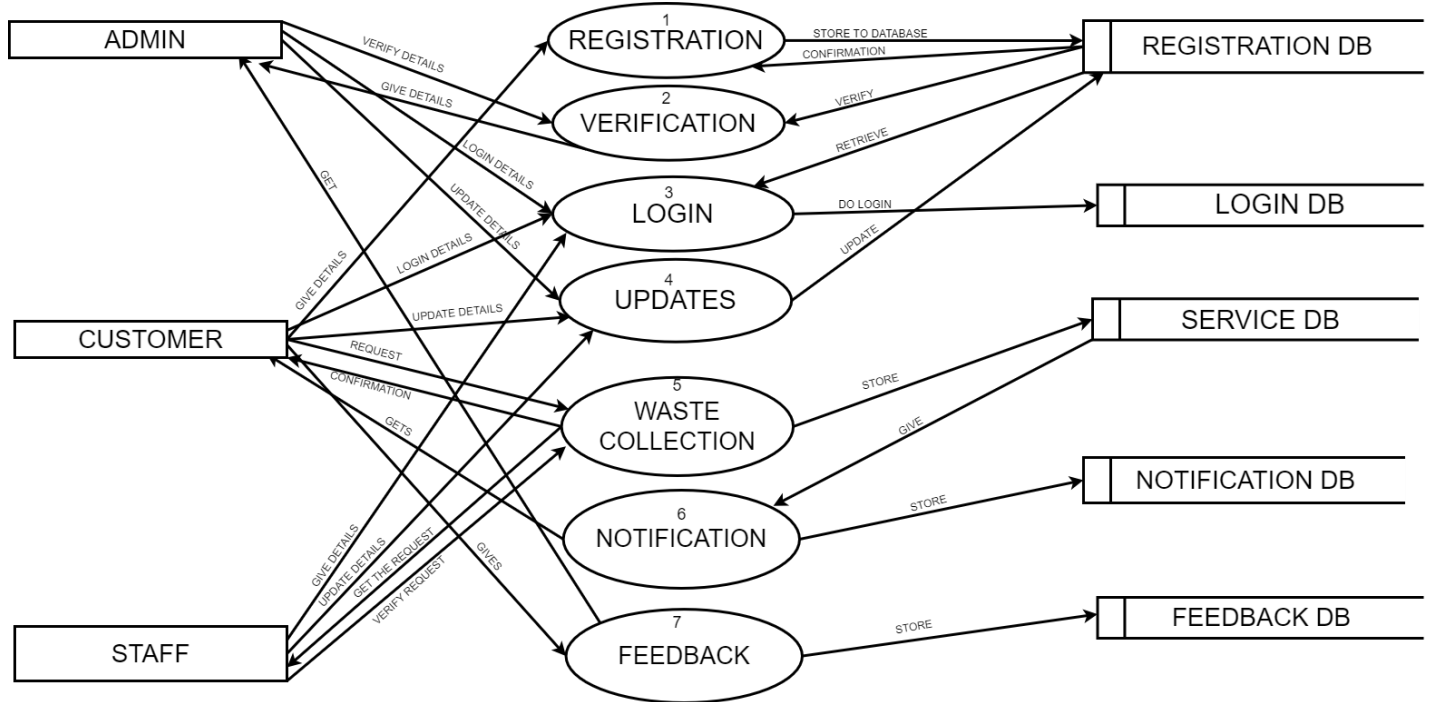


Image 4.1.3.1 Level 2 DFD (5)



4.2 ER DIAGRAM

An Entity Relationship (ER) Diagram is a type of flowchart that illustrates how “entities” such as people, objects or concepts relate to each other within a system.

Classifying Relationships

Relationships are classified by their degree, connectivity, cardinality, direction, type and existence.

Degree of Relationships

The degree of a relationship is the number of entities associated with the relationship. The n-ary relationship is there general form for degree n. Special cases are binary, ternary where the degree is 2 and 3 respectively.

Connectivity and Cardinality

The connectivity of a relationship describes the mapping of associated entity instances in the relationship. The values of connectivity are “one” or “many”. The cardinality of a relationship is the actual number of related occurrences for each of the two entities. The basic types of connectivity of relations are: One-to-one, one-to-many, many-to-many.

One-to-One (1:1)

One-to-many (1: N)

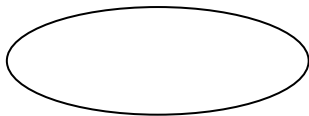
Many-to-many (M: N)

The symbols used ER Diagram is

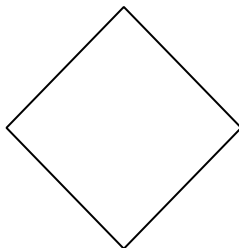
Entity



Attributes



Relationship



Lines



ER DIAGRAM FOR ECOBIN

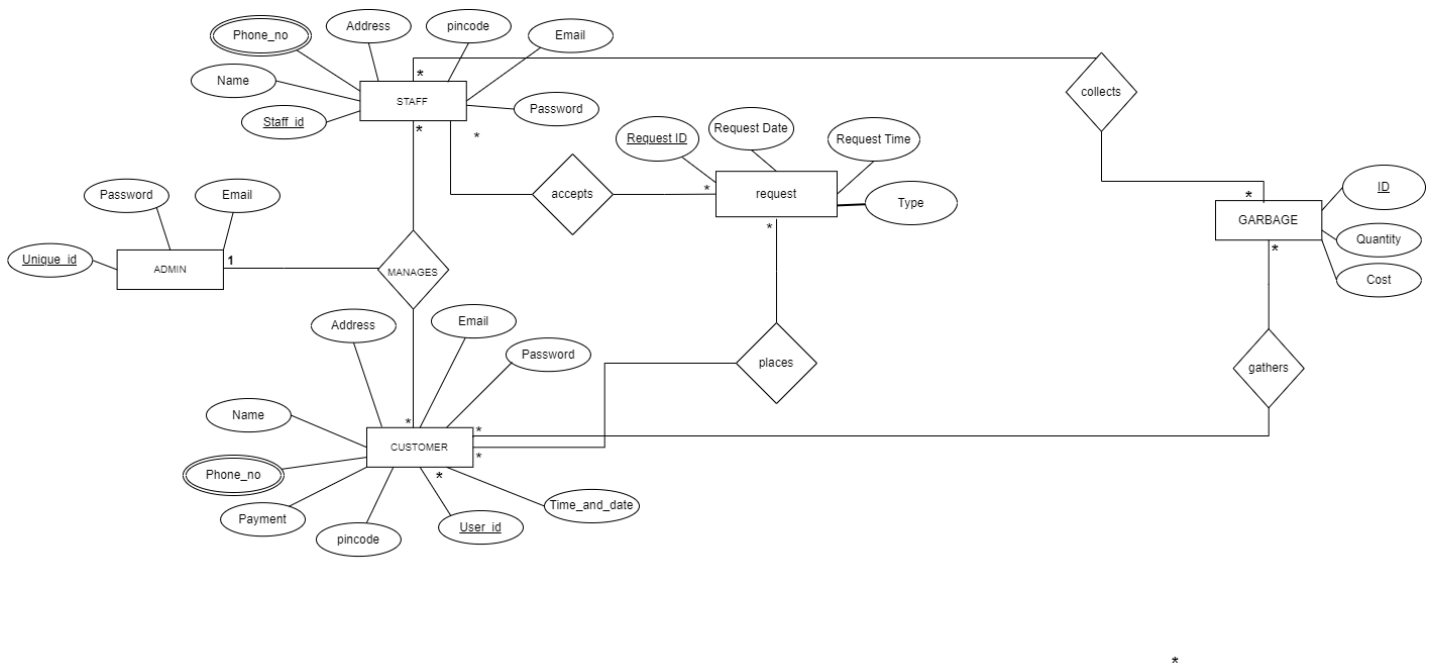


Image 4.2.2 ER DIAGRAM

The ER diagram illustrates the entities and their corresponding attributes in the waste collection and e-commerce system.

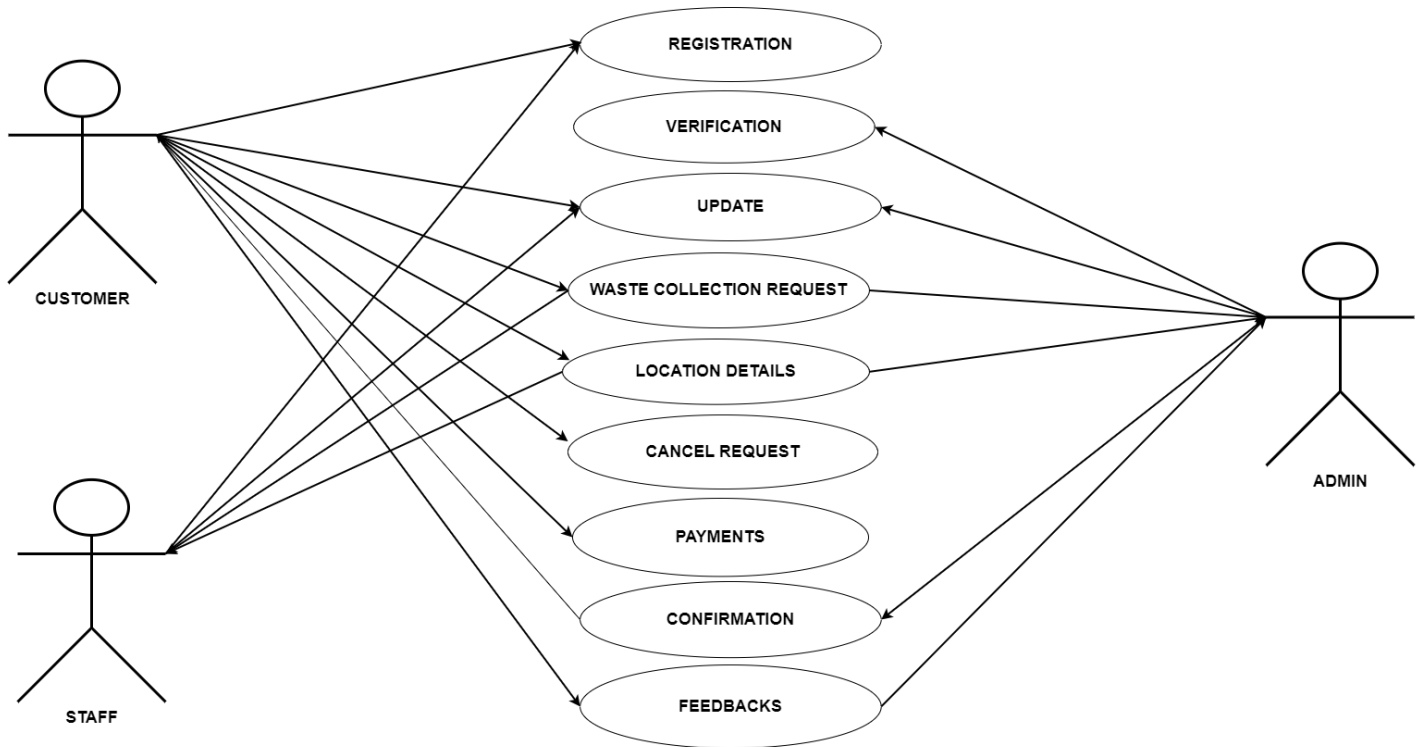
The main entities include Product, Staff, Admin, Delivery Guy, Customer, Payment, Garbage, Service, and Order.

Each entity has its unique identifier attribute

4.3 USER Interface Diagram

A **use case diagram** is a graphical depiction of a user's possible interactions with a system. A use case diagram shows various use cases and different types of users the system has and will often be accompanied by other types of diagrams as well. The use cases are represented by either circles or ellipses. The actors are often shown as stick figures.

USE CASE



4.3 Database Design

TABLE:1

Table name: staff_login

Table description: staff details

it contains the staff details.

Field Name	Data Type	Constraints
id	int (11)	PRIMARY KEY
name	varchar (255)	NOT NULL
pincode	int (11)	NOT NULL
email	varchar (255)	UNIQUE
Password	varchar (255)	NOT NULL
address	varchar (22)	NOT NULL
phone	int (11)	NOT NULL

TABLE:2

Table name: waste_request_confirm

Table description: waste request conformation

it contains waste request conformation details for the customer.

Field Name	Data Type	Constraints
Id	int (11)	Primary key
name	varchar (255)	Not null
address	varchar (255)	Not null
date	date	check
time	time	default

Table :3

Table name: waste_request

Table description: it contains waste request details given by the customer.

Field Name	Data Type	Constraints
id	int (11)	Primary key
type	varchar (10)	Not null
date	date	check
time	time	default
email	varchar (255)	unique
name	varchar (255)	Not null
pincode	int (11)	check
phone	int (11)	check

address	varchar (255)	Not null
confirm	varchar (15)	check
description	varchar (255)	check

Table :4

Table name:login

Table description:login details

it contains the login details.

Field Name	Data Type	Constraints
Id	int (11)	Primary key
Email	varchar (30)	unique
Password	varchar (50)	check

Table 5

Table name: register

Table description: register details

it contains register details of the customers.

Field Name	Data Type	Constraints
Id	int (11)	Primary key
Full Name	varchar (255)	Not null
AadhaarNumber	int (13)	check
Email	varchar (255)	unique
PhoneNo	int (11)	check
Password	varchar (255)	check

Address	varchar (255)	Not null
Gender	varchar (10)	Not null
Aadharproof	longblob	check
Pincode	int (15)	Not null
State	varchar (30)	Not null
Otp	varchar (11)	check

CHAPTER 5

CODING AND IMPLEMENTAION

5.1 Coding

Code tells a computer what actions to take and writing code is like creating a set of instructions. We have implemented a number of features through this website.

5.2 Implementation

```
<?php
error_reporting(0);
    include "register.php";
?>
<?php include "login.php"; ?>

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <meta http-equiv="X-UA-Compatible" content="ie=edge">
    <!-- Bootstrap CSS -->
    <link rel="stylesheet" href="../css/bootstrap.min.css">

    <!-- Font Awesome CSS -->
    <link rel="stylesheet" href="../css/all.min.css">

    <!-- Unicons -->
    <link rel="stylesheet" href="https://unicons.iconscout.com/release/v4.0.0/css/line.css" />

    <!-- Google Font -->
    <link href="https://fonts.googleapis.com/css?family=Ubuntu&display=swap" rel="stylesheet">

    <!-- Custom CSS -->
    <link rel="stylesheet" href="../css/style.css">

    <title>ECOBIN</title>
</head>
<body>
<section id="home" class="home">

    <div class="form_container">
        <i class="uil uil-times form_close"></i>
        <!-- Login From -->
        <div class="form login_form">
            <form action="" method="post" enctype="multipart/form-data">
```

```

<h2>Login</h2><span style="color: red;"><?php echo $login_error; ?></span>

<div class="input_box">
    <input type="email" name="email" placeholder="Enter your email" required />
    <i class="uil uil-envelope-alt email"></i>
</div>
<div class="input_box">
    <input type="password" name="password" placeholder="Enter your password"
required />
    <i class="uil uil-lock password"></i>
    <i class="uil uil-eye-slash pw_hide"></i>
</div>

<div class="option_field">
    <span class="checkbox">
        <input type="checkbox" id="check" />
        <label for="check">Remember me</label>
    </span>

    <a href="#" class="forgot_pw" id="forgot">Forgot password?</a>

</div>

<button class="button" name="login" >Login Now</button>

<div class="login_signup">Don't have an account? <a
href=" ../registration/register.html" >Registration</a></div>
</form>

</div>

<!-- Forgot From -->
<div class="form forget-form">
    <form action="" method="post" enctype="multipart/form-data" >
        <h2>Forgot password</h2>

        <div class="input_box">
            <input type="email" name="email" placeholder="Enter your email" required />
            <i class="uil uil-envelope-alt email"></i>
        </div>
        <!-- popup -->

        <button class="button" name="forgotemail" >Sent mail</button>

    </form>

</div>

```

```

    </div>
</section>

<!-- Start Navigation -->
<header>
    <nav class="navbar navbar-expand-sm pl-30 fixed-top navbarc">
        <a href="index.php" class="navbar-brand">Eco<span class="bin">Bin</span></a>
        <button type="button" class="navbar-toggler" data-toggle="collapse" data-
target="#myMenu">
            <span class="navbar-toggler-icon"></span>
        </button>
        <div class="collapse navbar-collapse" id="myMenu">
            <ul class="navbar-nav pl-5 custom-nav">
                <li class="nav-item"><a href="index.php" class="nav-link">Home</a></li>
                <li class="nav-item"><a href="#Services" class="nav-link">Services</a></li>
                <li class="nav-item"><a href="#registration" class="nav-link">Registration</a></li>
                <!-- <li class="nav-item"><a href="#home" class="nav-link" id="form-
open">Login</a></li> -->
                <li class="nav-item"><a href="#Contact" class="nav-link">Contact</a></li>
            </ul>

            <div class="h-right">
                <a href="#Follow">Follow Us</a>

                <a href="https://www.facebook.com/EcoBin" target="_blank" class="pr-2 fi-color"><i
class="fab fa-facebook-f"></i></a>
                <a href="https://twitter.com/EcoBin858" target="_blank" class="pr-2 fi-color"><i
class="fab fa-twitter"></i></a>
                <a href="#" target="_blank" class="pr-2 fi-color"><i class="fab fa-youtube"></i></a>
                <a href="#" target="_blank" class="pr-2 fi-color"><i class="fab fa-google-plus-
g"></i></a>
                <a href="#" target="_blank" class="pr-2 fi-color"><i class="fas fa-rss"></i></a>

            </div>
        </div>
    </nav>
</header>
<!-- end of navigation -->
<!-- Start Header Jumbotron-->
<header class="jumbotron back-image" >
    <div class="myclass">
        <h5 class="weltext">JOIN THE MOVEMENT TO SHAPE A CLEANER</h5>
        <h1 class="aim">Greener<br>Tomorrow</h1>
        <a href="#home" id="form-open" class="btn-l">Login</a>
        <a href="#registration" class="btn-s">Sign Up</a>
    </div>
</header>
<!-- End Header Jumbotron-->

<!-- Start Introduction Section -->
<div class="container img">

```

```

<div class="image">
  
</div>
<div class="SUCHITWA">

  <h3 class="text-center text-uppercase text-h3">Our aim</h3>
  <h2 class="text-h2">A step to <br>improve<br> Sucithwa<br>Mission</h2>
  <p>
    Our website aims to promote sustainable waste managementby facilitating digitalized
    collection and disposal methods,supporting "SUCHITWA MISSION" and contributing to a cleaner
    environment in kerala
  </p>
</div>
</div>
<!-- End Introduction Section -->

<!-- Start Registration Form -->
<!-- <?php //include('UserRegistration.php') ?> (REPLACE THE BELOW FORM WITH THIS CODE WHEN
THE USERREGISTRATION.PHP IS READY) -->

<div class="reg container " id="registration" >
  <h2 class="text-center acc">Create an Account</h2>
  <div class="row mt-4 mb-4">
    <div class="col-md-6 offset-md-3">
      <form action="" class="shadow-lg p-4" method="POST" enctype="multipart/form-data">
        <div class="form-group">
          <i class="fas fa-user icon"></i> <label for="name" class="font-weight-bold pl-2">Full
Name</label><input type="text" class="form-control" placeholder="Name"
name="fullName"required>
        </div>

        <div class="form-group">
          <i class="fas fa-user icon"></i>
          <label for="email" class="font-weight-bold pl-2">Email</label><span><?php echo
"<nav style='color:red;'">$errone</nav>"; ?></span>
          <input type="email" class="form-control" placeholder="Email" name="email"
required>
          <small class="form-text warn">We'll never share your email with anyone
else.</small>
        </div>

        <div class="form-group">
          <i class="fas fa-key icon"></i>
          <label for="pass" class="font-weight-bold ">Password</label>
          <br>
          <input type="password" class="form-" placeholder="Password"
name="password"required>
        </div>

```

```

<div class="form-group">
<i class="fas fa-phone icon"></i>
<label for="gender" class="font-weight-bold pl-2">gender</label>
<select class="form-control" name="gender" id="gender">
  <option value="male">Male</option>
  <option value="female">Female</option>
  <option value="others">Others</option>
</select>
</div>

<div class="form-group">
<i class="fas fa-phone icon"></i> <label for="phon" class="font-weight-bold pl-2">Phone
number</label><input type="number" class="form-control" placeholder="Phone
number"pattern="[0-9]{10}" name="phoneNumber" required>
</div>

<div class="form-group">
<i class="fas fa-home icon"></i> <label for="address" class="font-weight-bold pl-
2">Address</label><input type="text" class="form-control" placeholder="address line 1"
name="aaddress"required>
</div>

<div class="form-group">
<i class="fas fa-phone icon"></i> <label for="phon" class="font-weight-bold pl-
2">Aadhar number</label><input type="number" class="form-control" placeholder="Aadhar
number"pattern="[0-9]{12}" name="aadhaar-no" required><span><?php echo "<nav
style='color:red;'">$errora</nav>"; ?></span>
</div>

<div class="form-group">
<i class="fas fa-home icon"></i> <label for="pincode" class="font-weight-bold pl-
2">pincode</label><br><input type="number" class="form-" placeholder="PINCODE" pattern="[0-
9]{6}" id="pincode" name="pincode" required>
  <a class="btn-m" onclick="get_details()">CHECK</a>
</div>

<div class="form-group">
<i class="fas fa-home icon"></i> <label for="city" class="font-weight-bold pl-
2">city</label><br><input type="text" class="form-" placeholder="city" id="city" name="city"
readonly>
</div>

<div class="form-group">
<i class="fas fa-home icon"></i> <label for="Local area" class="font-weight-bold pl-
2">Local Area</label><br><input type="text" class="form-" id="localarea"
placeholder="Localarea" name="localarea" readonly >
</div>

<div class="form-group">

```

```

        <i class="fas fa-home icon"></i> <label for="proof" class="font-weight-bold pl-2">Aadhar proof (pdf)</label><br><input type="file" class="form-" name="aadhaarproof" accept=".pdf">
    </div>

    <!-- <div class="form-group">
        <label for="resume">Aadhar proof (PDF)</label>
        <input type="file" id="resume" name="resume" accept=".pdf" required>
    </div> -->

    <button type="submit" class="btn mt-5 btn-block shadow-sm font-weight-bold btn-m" name="register">Sign Up</button>
    <em style="font-size:10px;">Note - By clicking Sign Up, you agree to our Terms, Data Policy and Cookie Policy</em>
    <?php if(isset($regmsg)) {echo $regmsg;} ?>    <!-- PLEASE CHECK IF THIS IS CORRECT -->
</form>
</div>
</div>
</div>
<!-- End Registration Form -->

<!-- Start Contact US -->
<div class=" reg container" id="Contact">

    <di class="row">
    <div class="col-md-8">
    <!-- Start 1st Column -->
    <!-- <?php //include('contactform.php') ?> (ADD THIS WHEN YOU CREATE CONTACTFORM.PHP) -->

    <h2 class="contactus text-center mb-4">Contact Us</h2>
    <form action="" method="POST" enctype="multipart/form-data">
        <input type="text" class="form-control" name="name" placeholder="Name"><br>
        <input type="text" class="form-control" name="subject" placeholder="Subject"><br>
        <input type="email" class="form-control" name="email" placeholder="Email"><br>
        <textarea class="form-control" name="message" placeholder="How can we help you?" style="height:150px;"></textarea><br>
        <input type="submit" class="btn btn-primary btn m" name="contact-submit"><br><br>
    </form>
    </div>
    <!-- End 1st Column -->
    <div class="contact text-center"> <!-- Start 2nd Column -->
        <strong>Headquarter:</strong><br>
        EcoBin pvt Ltd,<br>
        Subash Nagar, delhi<br>
        New Delhi - 110064<br>
        Phone: +91 9871****22<br>
        <a href="#" target="_blank">www.EcoBin.com</a><br>
        <br> <br>
        <strong>Branch:</strong><br>
        EcoBin pvt Ltd,<br>
        karunagappalli, kollam<br>

```

```

kerala - 690542<br>
Phone: +91 9544****53<br>
<a href="#" target="_blank">www.EcoBin.com</a><br>
</div> <!-- End 2nd Column -->
</div>
</div>

<!-- End Contact US -->
<!-- Start Footer -->
<footer class="reg container-fluid bg-dark text-white">
  <div class="container">
    <div class="row py-3">
      <div class="col-md-6"> <!-- Start 1st Column -->
        <span class="pr-2" id="Follow">Follow Us: </span>
        <a href="#" target="_blank" class="pr-2 fi-color"><i class="fab fa-facebook-f"></i></a>
        <a href="#" target="_blank" class="pr-2 fi-color"><i class="fab fa-twitter"></i></a>
        <a href="#" target="_blank" class="pr-2 fi-color"><i class="fab fa-youtube"></i></a>
        <a href="#" target="_blank" class="pr-2 fi-color"><i class="fab fa-google-plus-
g"></i></a>
        <a href="#" target="_blank" class="pr-2 fi-color"><i class="fas fa-rss"></i></a>
      </div> <!-- End 1st Column -->
      <div class="col-md-6 text-right"> <!-- Start 2nd Column -->
        <small>Minor Project(Group 5)<br>Bsc Computer Science<br>Batch 2021-2024<br> &copy;
2023</small>
        <small class="ml-2"><a href="../staff/login.php">Staff Login</a></small>
      </div> <!-- End 2nd Column -->
    </div>
  </div>
</footer>

<!-- javascript starts here -->

<!-- checking PINCODE -->

<script >

function get_details() {
  var pincode = jQuery('#pincode').val();
  if (pincode == '') {
    jQuery('#city').val('');
    jQuery('#localarea').val('');
  } else {
    jQuery.ajax({
      url: 'get.php',
      type: "post",
      data: 'pincode=' + pincode,
      success: function (data) {
        if (data == 'no') {

```

```

        alert('Wrong Pincode');
        jQuery('#city').val('');
        jQuery('#localarea').val('');
    } else {
        try {
            var getData = $.parseJSON(data);
            jQuery('#city').val(getData.city); // Populate city input field
            jQuery('#localarea').val(getData.localarea); // Populate state input
field
        } catch (e) {
            console.error("Error parsing JSON: " + e);
        }
    }
    console.log(data);
}
});
}
}

</script>

<script src="https://code.jquery.com/jquery-3.6.0.min.js"></script>
<script src="../../js/jquery.min.js"></script>
<script src="../../js/popper.min.js"></script>
<script src="../../js/bootstrap.min.js"></script>
<script src="../../js/all.min.js"></script>
<script src="../../js/script.js"></script>
<script src="../../js/popup-login.js"></script>
</body>
</html>
<?php
    if(isset($_POST['forgotemail'])){
        include "forgot-password.php"; } ?>
<?php
    if(isset($_POST['register'])){
        include "register.php"; } ?>
<?php
    if(isset($_POST['contact-submit'])){
        include "contact-us.php"; } ?>

```

CHAPTER 6

TESTING

6.1 Higher Level Test Items

- Login
- Register
- Forgot password
- Email

6.2 Lower-Level Test Items

- Personal details of user
- Updation
- Data entering
- Data verification
- Registration

6.3 Unit Testing

In computer programming unit testing is a software verification and validation method in which a programmer tests if individual unit of source code are fit for use. A smallest testable part of an application in procedural programming in a unit may be an individual functions or procedure. Unit testing focus on the smallest unit of software design, the module. This is also known as “Module Testing”. The modules are tested separately. This testing is carried out during the programming stage itself. In this step each modules found to be working satisfactorily. The forms in this project are tested one by one and errors are corrected.

- Registration process of a user, staff.
- Login process of the user, staff and admin.
- Checked if the website can be load without any issues

6.4 Integration testing

Integration testing checks how different modules or components of ECOBIN work together and integrate with each other. This type of testing ensures that all the components of the platform work seamlessly and deliver the expected results. Test cases are designed to test the integration of various modules, and the tests are executed

to verify that the modules work correctly together. This type of testing can be performed manually, but automated testing tools are commonly used to ensure that all the integrations work correctly.

6.5 Functional Testing

Each item sets are tested in this phase of testing. The main goal is to ensure that no functions were missed out. For the smooth functioning of the website, functional testing is much necessary. Functionality testing checks if ECOBIN meets its functional requirements and specifications. It ensures that each function or feature of the platform works correctly and delivers the expected results. Test cases are designed for each feature or function of the platform, and the tests are executed to verify that the feature works as expected. This type of testing is usually performed manually, but automated testing tools can also be used to speed up the process.

CHAPTER 7

RESULTS AND DISCUSSIONS

7.1 User Module

The Users can create an account by providing their basic information such as name, email, and password. Users can view and update their profile where they can showcase their personal information such as name, address, phone number, and Email. Users can request services they need. User can monitor the service request and get the information regarding who when the service will be executed. Users can access various waste management techniques tips and trick which is provided by our system. The user can give feedback about the staff and their efficiency through our feedback form.

7.2 Admin Module

Admin is responsible for verifying user registration and granting admission to accepted user, updating staff details, entering staff details and registering them, responding to complaints and feedback.

7.3 Staff Module

Staff members have unique login credentials (username and password) to access the system. This ensures that only authorized personnel can make changes to pick up schedules. Upon logging in, staff members would be presented with a dashboard that displays the current and upcoming pickup requests, along with relevant details such as the location, type of waste, and scheduled pickup time. Staff members can update pickup schedules by modifying the date and time of upcoming pickups. This can be done manually by selecting the specific request and editing the relevant fields. Staff members can update pickup schedules by modifying the date and time of upcoming pickups. This can be done manually by selecting the specific request and editing the relevant fields.

CHAPTER 8

CONCLUSION AND FUTURE ENHANCEMENT

8.1 CONCLUSION

Users can easily register to the page and login to view various details. There is an admin to confirm the registration of users, manage their details and update the information that can view by the users. ECOBIN is very useful for Plastic waste collection and management. This application is designed for effective and smooth functioning for plastic waste collection. This particular project deals with the problems on managing a plastic waste collection and avoids the problems which occur when carried manually.

8.2 Future Enhancement

Future enhancements for an organization or platform like "Ecobin" could involve incorporating innovative technologies and strategies to further improve waste management, sustainability, and environmental conservation efforts. While I don't have specific knowledge of the developments after September 2021, here are some potential future enhancements that could be considered:

1. **IoT Integration:** Implementing Internet of Things (IoT) devices into waste bins could enable real-time monitoring of fill levels. This data can be used to optimize collection routes, reduce unnecessary pickups, and save fuel and resources.
2. **Smart Routing:** Utilize advanced route optimization algorithms to plan collection routes based on real-time data, minimizing travel distance and time while reducing carbon emissions.
3. **Data Analytics:** Develop sophisticated data analytics tools to analyse waste generation patterns, identify trends, and provide insights for more effective waste management strategies.
4. **Artificial Intelligence (AI) Sorting:** Integrate AI-powered sorting systems at recycling facilities to improve the accuracy and efficiency of sorting various types of recyclables.
5. **Waste-to-Energy Conversion:** Explore technologies for converting certain types of waste into energy through methods like anaerobic digestion, gasification, or waste-to-energy incineration.
6. **User-Friendly Mobile App:** Create a user-friendly mobile app for customers to easily schedule pickups, track their waste generation, and receive notifications about collection updates.
7. **Blockchain for Transparency:** Implement blockchain technology to track and verify the entire waste management process, ensuring transparency, traceability, and accountability.
8. **Educational Resources:** Enhance educational initiatives by providing digital resources, interactive tools, and virtual tours to educate the public about waste reduction, recycling, and sustainability.
9. **Augmented Reality (AR) Tools:** Develop AR applications that allow users to visualize the impact of their waste management choices, encouraging behaviour change.
10. **Community Engagement Platforms:** Build online platforms for communities to share ideas, success stories, and collaborate on local environmental projects.
11. **Carbon Footprint Tracking:** Integrate carbon footprint calculation tools to help organizations and individuals understand the environmental impact of their waste and consumption habits.
12. **Green Packaging Solutions:** Collaborate with businesses to explore and implement sustainable packaging alternatives to reduce waste at the source.

13. **Extended Producer Responsibility (EPR) Integration:** Partner with manufacturers to implement EPR programs where they take responsibility for the disposal and recycling of their products.
14. **Eco-Incentive Programs:** Develop reward systems that incentivize users to participate in waste reduction and recycling initiatives, earning points or discounts.
15. **Waste Minimization Strategies:** Offer consulting services to organizations to help them minimize waste generation through process improvements, packaging redesign, and efficient resource use.
16. **Virtual Reality (VR) Experiences:** Create VR simulations to immerse users in scenarios depicting the impact of waste mismanagement on the environment, fostering empathy and understanding.

These are just a few possibilities for future enhancements that "Ecobin" or a similar organization could consider. The key is to leverage technology, innovation, and community engagement to create a more sustainable and environmentally friendly future.

REFERENCES

- <https://aakri.in/>
- <https://www.wm.com/>
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